

Coding & Solution

Date	Team ID	Project Name	Maximum Marks
		SB Stocks	4

Key code implementations and solutions.

Milestone-wise Implementation:

Milestone 1: User Authentication

- Implemented user registration with email validation
- Password hashing

Milestone 2: Stock Market Data Integration

- Alpha Vantage API integration for real-time stock quotes
- Historical price retrieval

Milestone 3: Trading Engine

- Buy order processing: validate balance, deduct funds, add to portfolio
- Sell order processing

Milestone 4: Portfolio & Dashboard

- Real-time portfolio value calculation
- Profit/Loss track

Challenges & Solutions

S.No	Challenge	Solution Implemented
1	API rate limiting on Alpha Vantage free tier	Implemented caching layer with 60-second TTL
2	Real-time portfolio value updates	Batch API calls with async processing
3	Concurrent trade execution consistency	Database transaction locking for order processing
4	Responsive UI across devices	Bootstrap 5 grid system with mobile-first approach